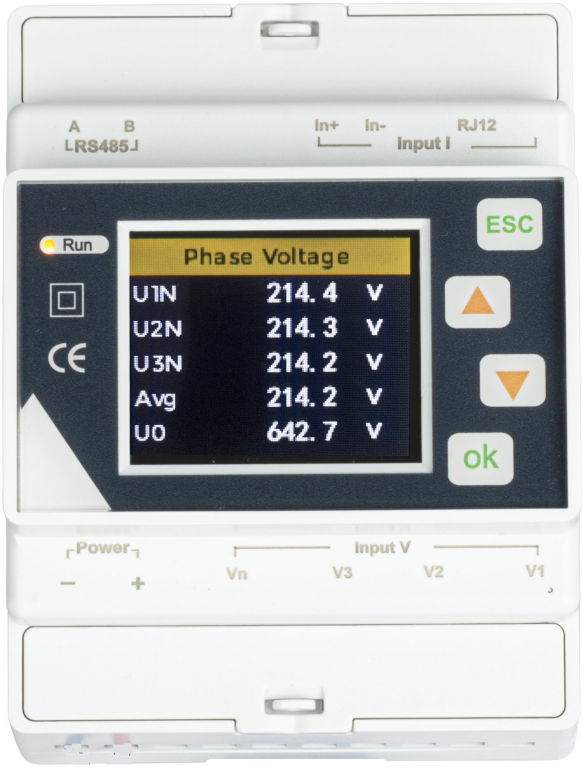
English

Modbus Register Table

S111



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1 Modbus Communication

Communication port	RS-485
Communication protocol	Modbus/RTU

The S111 adopts the standard Modbus/RTU protocol.

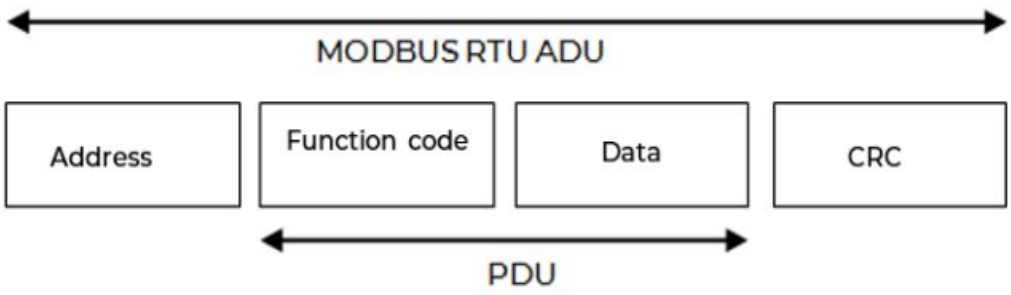
1.1 Modbus Communication settings

Before performing the Modbus/RTU communication, the following parameters need to be set via the meter interface.

Parameter	**Effective value**	**Default value**
Address	1–247	1
Baud rate	-1200 -2400 -4800 -9600 -19200 -38400 -57600 -115200	9600
Data bits	8	8
Parity check	– None – Odd – Even	None
Stop bit	1 – 2	1

1.2 Modbus/RTU data frame

Modbus/RTU data frame consists of 4 parts: Address field, Function code, Data, and Error verification.



1.3 PDU Request data Format

Function code	Data
8-Bits	N×8-Bits

1.4 Function code

Function codes are used to indicate how the device processes the instruction. The following table shows the available function codes and their descriptions.

Function code		Name of function code	Function
Decimal	Hexadecimal		
3	03H	Read holding register	Used to read meter's parameters
16	10H	Write multiple registers	Used to configure meter parameters

1.5 Register list

The register list has the following entries:

Register name	Register address	Operation Read/write	Register number	Type	Unit	Description

- Register name: used to indicate the purpose of the register.
- Register address: the address of Modbus register in decimal.
- Operation: used to indicate the operation that the register can perform.
  - R: read only
  - W: write
  - WC: write by register instruction
- Number of registers: indicates how many int16 sizes the register has.
- Type: describes the type of data.
- Unit: indicates the unit of the register value.
- Description: a description of the register.

1.5.1 Data type list

The following table lists the data types used in this document:

Type	Description	Range
UInt16	16-bit unsigned integer	0–65535
Int16	16-bit signed integer	-32768–+32767
UInt32	32-bit unsigned integer	0–4 294 967 295
UInt64	64-bit unsigned integer	0–18 446 744 073 709 551 615
UTF8	8-bit UTF	Multi-byte unicode coding
Float32	32-bit floating point number	Standard IEEE floating point numbers (single precision)
Date Time	Date and Time types	-
Time	Time type	-

Data Time format

Byte	Position															
	15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	Year (2000–2099)															
2	Month (1–12)								Day (1–31)							
3	Hour (0–23)								Second (0–59)							
4	Second (0–59)															

Time format

Byte	Position															
	15	**14**	**13**	**12**	**11**	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
1	Hour (0–23)															
2	Second (0–59)															
3	Second (0–59)															

1.5.2 Configure the device via the Modbus/RTU

You can use the function code 16 to write instructions to the device and configure the device parameters.

The device parameters can only be configured by writing the corresponding data to the "configuration instruction register", that is, writing the corresponding data to the address starting from 300 to configure the parameters.

Configuration results:

The configuration results can be obtained by reading registers 424 and 425.

Register address	Description	Size (UInt16)	Data (example)
424	Configuration instruction code	1	1001 (set Date Time)
425	Configuration results	1	0 = configuration successful 80 = invalid instruction code 81 = invalid parameter value 82 = number of invalid parameters 83 = instruction not executed

1.6 Modbus/RTU Function code

1.6.1 Function code (0x10=16) Operation Instructions

The function code (0x10=16) is used to configure parameters of the device, and its request and return instructions are defined as follows:

Data format of the request command:

No.	Name	Type	Range (Decimal)	Description
1	Device address	UInt8	1-247	
2	Function code	UInt8	16	
3	Register start address	UInt16	-	High byte first (sending sequence)
4	Number of configuration registers	UInt16	1-123	High byte first (sending sequence)
5	Data length	UInt8		Number of configuration registers * 2
6	First register configuration data	UInt16	-	High byte first (sending sequence)

7	…	UInt16	-	High byte first (sending sequence)
8	Nth register configuration data	UInt16	-	High byte first (sending sequence)
9	CRC-16 parity code	UInt16	-	Low byte first (sending sequence)

Data format of the reture command:

No.	Name	Type	Range (Decimal)	Description
1	Device address	UInt8	1-247	
2	Function code	UInt8	16	
3	Register start address	UInt16	300	High byte first
4	Number of configuration registers	UInt16	1-123	High byte first
5	CRC-16 parity code	UInt16	-	Low byte first

Attention!

The function code (0x10=16) can only write data to the "configuration instruction register", that is, it can only write data to the register starting from the address 300.

For example: Configure the device time (command =1200, set to: 2022-11-1 12:20:00)

No.	Name	Type	Value (Decimal)	Value (HEX)	Description
1	Device address	UInt8	1	01	
2	Function code	UInt8	16	10	
3	Data byte length	UInt16	300	012C	Configuration register start address
4	Number of read registers	UInt16	7	0007	Configure Time. The instruction and parameter use a total of 7 registers.
5	Data length	UInt8	14	0E	Number of configuration registers * 2
6	Register 300 write value	UInt16	1200	04B0	Instruction code 1200 to configure time
7	Register 301 write value	UInt16	2022	07E6	Year of configuration time = 2022
8	Register 302	UInt16	11	000B	Month of configuration time = 11

	write value				
9	Register 303 write value	UInt16	1	0001	Day of configuration time = 1
10	Register 304 write value	UInt16	12	000C	Time of configuration = 12
11	Register 305 write value	UInt16	20	0014	Minutes of configuration time =20
12	Register 306 write value	UInt16	0	0000	Seconds of configuration time = 0
13	CRC-16 parity code	UInt16	35524	8AC4	low byte first(sending sequence)

The order of sending bytes is as follows:

01 10 01 2C 00 07 0E 04 B0 07 E6 00 0B 00 01 00 0C 00 14 00 00 C4 8A

After the configuration is successful, the received data packets are as follows:

01 10 01 2C 00 07 41 FE

No.	Name	Type	Value (Decimal)	Value (HEX)
1	Device address	UInt8	01	1
2	Function code	UInt8	10	16
3	Register start address	UInt16	012C	300
4	Number of configuration registers	UInt16	0007	7
5	CRC-16 parity code	UInt16	41FE	

1.6.2 Function code (0x03=3) operation instructions

The function code (0x03=3) is to read parameters of the device register. Its request data and return data format are as follows:

Request data format:

No.	Name	Type	Range (Decimal)	Description
1	Device address	UInt8	1-247	
2	Function code	UInt8	3	
3	Register start address	UInt16	-	High byte first (sending sequence)
4	Number of read registers	UInt16	1 – 125	High byte first (sending sequence)
5	CRC-16 parity code	UInt16	-	Low byte first (sending sequence)

Return data format:

No.	Name	Type	Range (Decimal)	Description
1	Device address	UInt8	1-247	
2	Function code	UInt8	3	
3	Data byte length	UInt8	-	Number of read registers * 2
4	1st register data		-	High byte first
5	…		-	High byte first
6	The nth register data		-	High byte first
7	CRC-16 parity code	UInt16	-	Low byte first

Example of reading device parameters:

Read the voltage values of A, B and C (the starting address of the voltage register is 1010):

No.	Name	Type	Range (Decimal)	Range (HEX)	Description
1	Device address	UInt8	1	0x01	
2	Function code	UInt8	3	0x03	
3	Register start address	UInt16	1010	0x03F2	
4	Number of read registers	UInt16	6	0x0006	

5	CRC-16 parity code	UInt16	32612	0x7F64	Low byte first (sending sequence)

The order of sending bytes is as follows:

01 03 03 F2 00 06 64 7F

The received packets are as follows:

01 03 0C 43 5C 00 00 43 5D 00 00 43 5E 00 00 14 AC

No.	Name	Type	Hex	Decimal
1	Device address	UInt8	01	1
2	Function code	UInt8	03	3
3	Number of bytes of reading register	UInt8	0C	12
4	Phase A voltage	float32	435C0000	220V
5	Phase B voltage	float32	435D0000	221V
6	Pphase C voltage	float32	435E0000	222V
7	CRC-16 parity code	UInt16	14AC	

1.6.3 Error response

Error response data format:

No.	Name	Type	Decimal	Hexadecimal	Description
1	Device address	UInt8	1-247	0x01-0xF7	
2	Function code	UInt8	(128+3) (128+16)	(0x80+0x03) (0x80+0x10)	
3	Error code	UInt8			
4	CRC-16 parity code	UInt16			Low byte first

Modbus Error code:

Code (HEX)	Name	Description
0x01	Illegal function code	The function codes 3 or 6 supported by the device is not used.
0x02	Illegal data address	The register data written or read is not a supported address range.
0x03	Illegal data value	The data value written to the register does not meet the requirements.
0x04	Device error	An unknown error occurred

1.7 List of configuration instructions

1.7.1 System parameter setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1001	W	1	UInt16	-	0, 1, 2, 3, 4, 5	Wiring mode 0=3P4W_4CT 1=3P4W_3CT 2=3P3W_3CT 3=3P3W_2CT 4=1P3W 5=1P2W
	W	1	UInt16	Hz	50, 60	Grid frequency
	W	1	UInt16	V	1-65535	Nominal voltage (not including VT ratio)
	W	2	UInt32	-	1-99999999	VT ratio, 10000 times magnification
	W	2	UInt32	-	1-99999999	CT ratio, 10000 times magnification

1.7.2 Parameter setting of A, B, C current transformers

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
0	W	1	UInt16	-	0, 1	Phase A, B, C current access mode 0 = Rogowski coil access 1 = VCT access
	W	2	UInt32	A	1 - 999999	Phase A, B, C Rogowski coil input value
	W	2	UInt32	mV@50Hz mV@60Hz	1 - 99999	Phase A, B, CRogowski coil output =Actual value*100 **Note**: The output should be set according to the set power grid frequency.
	W	2	UInt32	A	1 - 999999	Nominal current of phase A, B, C Rogowski coil
	W	2	UInt32	A	1 - 999999	Phase A, B, C VCT input value
	W	2	UInt32	mV	1 - 99999	Phase A, B, C VCT

						output =Actual value*100
	W	2	UInt32	A	1-999999	VCT nominal current of phase A, B, C

1.7.3 Parameter setting of N-phase current transformer

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1003	W	1	UInt16	-	0, 1	Phase N current access mode 0 = Rogowski coil access 1 = VCT access
	W	2	UInt32	A	1-999999	Phase N Rogowski coil input value
	W	2	UInt32	MV@ 50 Hz mV@ 60 Hz	1-99999	Phase N Rogowski coil output =Actual value*100 **Note**: The output should be set according to the set power grid frequency.
	W	2	UInt32	A	1-999999	Nominal current of phase N Rogowski coil
	W	2	UInt32	A	1-999999	Phase N VCT input value
	W	2	UInt32	mV	1-99999	Phase N VCT output =Actual value*100
	W	2	UInt32	A	1-999999	VCT nominal current of phase N

1.7.4 Direction setting of phase ABC current

When the coil direction is different from the actual one, you can use this configuration to change the current direction.

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1010	W	1	UInt16	-	0, 1	Current direction of phase A 0 = Forward (Default) 1 = Reverse
	W	1	UInt16	-	0, 1	Current direction of phase B 0 = Forward (Default) 1 = Reverse

	W	1	UInt16	-	0, 1	Current direction of phase C 0 = Forward (Default) 1 = Reverse

1.7.5 Channel setting of phase ABC current

When the current and voltage do not match, you can use this configuration to modify the current channel selection to make the phase of current and voltage consistent.

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1011	W	1	UInt16	-	0, 1, 2	Channel selection of phase A 0 = Current channel 1 (Default) 1 = Current channel 2 2 = Current channel 3
	W	1	UInt16	-	0, 1, 2	Channel selection of phase B 0 = Current channel 1 1 = Current channel 2 (Default) 2 = Current channel 3
	W	1	UInt16	-	0, 1, 2	Channel selection of phase C 0 = Current channel 1 1 = Current channel 2 2 = Current channel 3 (Default)

1.7.6 Zero drift suppression setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1020	W	1	UInt16	%	0~1000	Voltage zero drift suppression Take (nominal voltage * VT variable ratio) as a reference = Actual value * 100 Default: 10
	W	1	UInt16	%	0~1000	Current zero drift suppression Take (nominal current * VT variable ratio) as a reference = Actual value * 100 Default: 10

1.7.7 Demand parameter setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1060	W	1	UInt16	-	0, 1	Demand calculation method 0= fixed, 1= sliding type
	W	1	UInt16	minute	01/01/60	Demand calculation interval

1.7.8 Tariff mode setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1070	W	1	UInt16	-	0, 1	Tariff switch mode 0 = Manual switch 1 = RTC switch

1.7.9 Manual tariff setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1071	W	1	UInt16	-	0 - 5	Manual tariff setting 1 = Tariff select 1 2 = Tariff select 2 … 5 = Tariff select 6 **Note**: This setting is only valid if the tariff mode is manual switching.

1.7.10 RTC tariff period setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1072	W	3	Time	-	-	Ta Start time
	W	3	Time	-	-	Tb Start time
	W	3	Time	-	-	Tc Start time
	W	3	Time	-	-	Td Start time
	W	3	Time	-	-	Te Start time

	W	3	Time	-	-	Tf Start time

1.7.11 RTC tariff select setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1073	W	1	UInt16	-	0 - 5	Ta Tariff select 0 = Tariff 1 1 = Tariff 2 … 5=Tariff 6
	W	1	UInt16	-	0 - 5	Tb Tariff select
	W	1	UInt16	-	0 - 5	Tc Tariff select
	W	1	UInt16	-	0 - 5	Td Tariff select
	W	1	UInt16	-	0 - 5	Te Tariff select
	W	1	UInt16	-	0 - 5	Tf Tariff select

1.7.12 Device time setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1200	W	1	UInt16	-	2000-2099	Year
	W	1	UInt16	-	1-12	Month
	W	1	UInt16	-	1-31	Date
	W	1	UInt16	-	0-23	Hour
	W	1	UInt16	-	0-59	Minute
	W	1	UInt16	-	0-59	Second

1.7.13 Communication parameter setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1210	W	1	UInt16	-	1-247	Slave address
	W	1	UInt16	-	0-6	Baud rate 0 = 2400 1 = 4800 2 = 9600 3 = 19200 4 = 38400 5 = 57600 6 = 115200
	W	1	UInt16	-	0, 1, 2	Parity check 0 = none check 1 = odd check 2 = even check
	W	1	UInt16	-	1, 2	Stop bit 1 = 1bit 2 = 2bit

1.7.14 Reset setting

Instruction code	R/W	Size	Type	Unit	Range (Decimal)	Description
1301	W	1	UInt16	-	1-5	1: Reset Max. and Min. 2: Reset Demand Max 3: Reset Tariff Energy 4: Reset Energy 5: Reset ALL

1.8 Register list

The register list has the following headings:

Register alias	Register address	Pperation Read/Write	Size	Type	Unit	Description

- Register alias: Used to refer to the meaning of registers
- Register address: The initial address of Modbus communication register is in decimal format, and the address is the real address without offset.
- Operation: Indicates the operation that the register can perform.
  - R: readable
  - W: It can be written directly through function code 16
  - WC: the current register needs to be configured indirectly by writing configuration data to the address starting from the instruction register 300.
- Size: Indicates how many Modbus registers are occupied. One Modbus register is 16 bit.
- Type: For the type of data code, see the Data type table
- Unit: Unit of register value
- Description: the function of this register.

Data type table

Type	Description	Range
UInt16	Unsigned 16 bits integer	0 ~ 65535
Int16	Signed 16 bits integer	-32768 ~ +32767
UInt32	Unsigned 32-bit integer	0 ~ 4294967295
UInt64	Unsigned 64 bits integer	0 ~ 18446744073709551615
Int64	Signed 64 bits integer	-9223372036854775808 ~ 9223372036854775808
UTF8	8-bit UTF code	Multibyte Unicode encoding
Float32	32-bit floating point	Standard IEEE floating point data (single precision)
Date Time	Time type	-

Data Time format

Byte	Position														
	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
1	Year(2000–2099)														
2	Month (1–12)								Date (1 – 31)						
3	Hour (0–23)								Minute (0 – 59)						
4	Second (0–59999)														

1.9 Modbus Register List

1.9.1 Equipment parameters

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Device model	60	R	10	UTF8	-	
Serial No.	70	R	2	UInt3 2	-	
APP Version No.	72	R	1	UInt1 6	-	Format: X.Y
Date and time	75	R/WC	4	Date time	-	Reg.75: Year 2000-2099 Reg.76: Month (b15:b8), Date (b7:b0) Reg. 77: Hour (b15:b8) , Minute (b7:b0) Reg. 78: Second

1.9.2 Communication parameter

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Slave address	80	R/WC	1	UInt16	-	1-247
Baud rate	81	R/WC	1	UInt16	-	0=2400, 1=4800 2=9600, 3=19200 4=38400, 5=57600 6=115200
Parity	82	R/WC	1	UInt16	-	0 = None 1 = Odd, 2 = Even
Stop bit	83	R/WC	1	UInt16	-	0 = 1 bit, 1 = 2 bit

1.9.3 Voltage and current phase sequence

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Voltage current phase sequence state	220	R	1	UInt16	1 2 - 3 4	0 = voltage sequence is correct, current sequence is correct 1 = voltage sequence wrong, ccurrent sequence correct 2 = voltage sequence correct, current sequence wrong 3 = voltage sequence wrong, current sequence wrong

Note: Current phase sequence may be errors when the current is less than 1% of the nominal current

1.9.4 Configure instruction register

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Instruction code	300	R/W	1	UInt16	-	
Instruction parameters 0 01	301	R/W	1	UInt16	-	
Instruction parameters 0 02	302	R/W	1	UInt16	-	
…	…	R/W	1	UInt16	-	
Instruction parameters 1 23	423	R/W	1	UInt16	-	
Configuration instruction code	424	R	1	UInt16	-	
Configuration results	425	R	1	UInt16	-	0 = valid operation 80 = invalid instruction code 81 = invalid instruction parameter 82 = number of invalid

						instruction parameters 83 = operation not executed

1.9.5 Power system

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Wiring mode	500	R/WC	1	UInt16	-	0=3P4W_4CT 1=3P4W_3CT 2=3P3W_3CT 3=3P3W_2CT 4=1P3W 5=1P2W
Grid frequency	501	R/WC	1	UInt16	Hz	
Nominal voltage	502	R/WC	1	UInt16	V	The VT ratio is not included.
VT Ratio	503	R/WC	2	UInt32	-	Actual value = read value /10000
CT Ratio	505	R/WC	2	UInt32	-	Actual value = read value /10000
Phase ABC current transformer						
Phase ABC current access type	510	R/WC	1	UInt16	-	0 = Rogowski coil 1 = VCT
Phase ABC Rogowski Coil Pri	511	R/WC	2	UInt32	A	
Phase ABC Rogowski Coil Sec	513	R/WC	2	UInt32	MV @ 50Hz mV @ 60Hz	Actual value = read value /100
Nominal current of Phase ABC Rogowski Coil	515	R/WC	2	UInt32	A	
Phase ABC VCT Pri	517	R/WC	2	UInt32	A	
Phase ABC VCT Sec	519	R/WC	2	UInt32	mV	Actual value = read value /100
Nominal current of	521	R/WC	2	UInt32	A	

Phase ABC VCT						
Phase-N current transformer						
Phase N current access type	530	R/WC	1	UInt16	-	0 = Rogowski coil 1 = VCT
Phase N Rogowski Coil Pri	531	R/WC	2	UInt32	A	
Phase N Rogowski Coil Sec	533	R/WC	2	UInt32	mV @50 Hz mV @60 Hz	Actual value = read value /100
Nominal current of Phase N Rogowski Coil	535	R/WC	2	UInt32	A	
Phase N VCT Pri	537	R/WC	2	UInt32	A	
Phase N VCT Sec	539	R/WC	2	UInt32	mV	Actual value = read value /100
Nominal current of Phase N VCT	541	R/WC	2	UInt32	A	

1.9.6 Current direction setting

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Current direction of phase A	550	R/WC	1	UInt16	-	Current direction of phase A 0 = Forward (Default) 1 = Reverse
Current direction of phase B	551	R/WC	1	UInt16	-	Current direction of phase B 0 = Forward (Default) 1 = Reverse
Current direction of phase C	552	R/WC	1	UInt16	-	Current direction of phase C 0 = Forward (Default) 1 = Reverse

1.9.7 Current channel selection

When the current and voltage do not match, you can use this configuration to modify the current channel selection to make the phase of current and voltage consistent.

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Current channel of phase A	553	R/WC	1	UInt16	-	Current channel of phase A 0 = Channel 1 (Default) 1 = Channel 2 2 = Channel 3
Current channel of phase B	554	R/WC	1	UInt16	-	Current channel of phase B 0 = Channel 1 1 = Channel 2 (Default) 2 = Channel 3
Current channel of phase C	555	R/WC	1	UInt16	-	Current channel of phase C 0 = Channel 1 1 = Channel 2 2 = Channel 3 (Default)

1.9.8 Zero drift suppression parameter

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Voltage zero drift suppression	600	R/WC	1	UInt16	%	Voltage zero drift suppression Take (nominal voltage *VT ratio) as a reference Actual value = Read value /100
Current zero drift suppression	601	R/WC	1	UInt16	%	Current zero drift suppression Take (nominal current *CT ratio) as reference Actual value = Read value /100

1.9.9 Tariff parameter

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Current tariff	800	R	1	UInt16	-	Current Tariff 0-5 = Tariff 1- Tariff 6
Tariff switching mode	801	R/WC	1	UInt16	-	Tariff switching mode 0=Manual 1=RTC
Manual tariff selection	802	R/WC	1	UInt16	-	Manual Tariff selection 0-5 = Tariff 1- Tariff 6
RTC Ta Start	803	R/WC	3	Time	-	RTC Ta Start
RTC Tb Start	806	R/WC	3	Time	-	RTC Tb Start
RTC Tc Start	809	R/WC	3	Time	-	RTC Tc Start
RTC Td Start	812	R/WC	3	Time	-	RTC Td Start
RTC Te Start	815	R/WC	3	Time		RTC Te Start
RTC Tf Start	818	R/WC	3	Time	-	RTC Tf Start
RTC Ta tariff select	821	R/WC	1	UInt16	-	Ta tariff select 0-5 = Tariff 1- Tariff 6
RTC Tb tariff select	822	R/WC	1	UInt16	-	Tb tariff select 0-5 = Tariff 1- Tariff 6
RTC Tc tariff select	823	R/WC	1	UInt16	-	Tc tariff select 0-5 = Tariff 1- Tariff 6
RTC Td tariff select	824	R/WC	1	UInt16	-	Td tariff select 0-5 = Tariff 1- Tariff 6
RTC Te tariff select	825	R/WC	1	UInt16	-	Te tariff select 0-5 = Tariff 1- Tariff 6
RTC Tf tariff select	826	R/WC	1	UInt16	-	Tf tariff select 0-5 = Tariff 1- Tariff 6

1.9.10 Voltage, current, power, power factor

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Current						
IA	1000	R	2	Float32	A	Phase A current
IB	1002	R	2	Float32	A	Phase B current
IC	1004	R	2	Float32	A	Phase C current
Current Avg	1006	R	2	Float32	A	Average value of ABC three-phase current
IN	1008	R	2	Float32	A	Phase N current
Phase voltage						
UA	1010	R	2	Float32	V	UA-UN voltage
UB	1012	R	2	Float32	V	UB-UN voltage
UC	1014	R	2	Float32	V	UC-UN voltage
Phase Voltage Avg	1016	R	2	Float32	V	Average value of ABC three-phase phase voltage
U0	1018	R	2	Float32	V	Zero sequence voltage
Line voltage						
UAB	1020	R	2	Float32	V	UA-UB voltage
UBC	1022	R	2	Float32	V	UB-UC voltage
UCA	1024	R	2	Float32	V	UC-UA voltage
Line Voltage Avg	1026	R	2	Float32	V	Average value of three-phase line voltage
Active power						
PA	1028	R	2	Float32	kW	Phase A Active power
PB	1030	R	2	Float32	kW	Phase B Active power

PC	1032	R	2	Float32	kW	Phase C Active power
PTotal	1034	R	2	Float32	kW	Total Active power
Reactive power						
QA	1036	R	2	Float32	kVAR	Phase A Reactive power
QB	1038	R	2	Float32	kVAR	Phase B Reactive power
QC	1040	R	2	Float32	kVAR	Phase C Reactive power
QTotal	1042	R	2	Float32	kVAR	Total Reactive power
Apparent power						
SA	1044	R	2	Float32	kVA	Phase A Apparent power
SB	1046	R	2	Float32	kVA	Phase B Apparent power
SC	1048	R	2	Float32	kVA	Phase C Apparent power
STotal	1050	R	2	Float32	kVA	Total Apparent power
Power factor						
PFA	1052	R	2	Float32	-	Phase A Power factor
PFB	1054	R	2	Float32	-	Phase B Power factor
PFC	1056	R	2	Float32	-	Phase C Power factor
PFTotal	1058	R	2	Float32	-	Total Power factor
Fundamental harmonic power factor						
DPFA	1060	R	2	Float32	-	Phase A Fundamental harmonic power factor
DPFB	1062	R	2	Float32	-	Phase B Fundamental harmonic power factor
DPFC	1064	R	2	Float32	-	Phase C Fundamental harmonic power factor
DPFTotal	1066	R	2	Float32	-	Total Fundamental harmonic power factor
Frequency						
FreqA	1068	R	2	Float32	Hz	Phase A Frequency

FreqB	1070	R	2	Float32	Hz	Phase B Frequency
FreqC	1072	R	2	Float32	Hz	Phase C Frequency
FreqTotal	1074	R	2	Float32	Hz	Total Frequency

1.9.11 Energy

The data types of electric energy are Int64 and UInt32. The units and sizes of the two types are different.

When the total electric energy reaches 1.0 x kWh, 1.0 x kVarh, or 1.0 x kVah, the electric energy of each phase will be cleared automatically.

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Active Energy-Int64						
EPAImp	2500	R	4	Int64	Wh	Phase A Positive active energy
EPBImp	2504	R	4	Int64	Wh	Phase B Positive active energy
EPCImp	2508	R	4	Int64	Wh	Phase C Positive active energy
EPImp	2512	R	4	Int64	Wh	Total Positive active energy
EPAExp	2516	R	4	Int64	Wh	Phase A Reverse active energy
EPBExp	2520	R	4	Int64	Wh	Phase B Reverse active energy
EPCExp	2524	R	4	Int64	Wh	Phase C Reverse active energy
EPExp	2528	R	4	Int64	Wh	Total Reverse active energy
Reactive energy-Int64						
EQAImp	2532	R	4	Int64	VARh	Phase A Positive reactive energy
EQBImp	2536	R	4	Int64	VARh	Phase B Positive reactive energy
EQCImp	2540	R	4	Int64	VARh	Phase C Positive reactive energy

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
EQImp	2544	R	4	Int64	VARh	Total Positive reactive energy
EQAExp	2548	R	4	Int64	VARh	Phase A Reverse reactive energy
EQBExp	2552	R	4	Int64	VARh	Phase B Reverse reactive energy
EQCExp	2556	R	4	Int64	VARh	Phase C Reverse reactive energy
EQExp	2560	R	4	Int64	VARh	Total Reverse reactive energy
Apparent Energy-Int64						
ESA	2564	R	4	Int64	VAh	Phase A Apparent Energy
ESB	2568	R	4	Int64	VAh	Phase B Apparent Energy
ESC	2572	R	4	Int64	VAh	Phase C Apparent Energy
ES	2576	R	4	Int64	VAh	Total Apparent Energy
UInt32 Energy						
Active Energy- UInt32						
EPAImp	2600	R	2	UInt32	kWh	Phase A Positive active energy
EPBImp	2602	R	2	UInt32	kWh	Phase B Positive active energy
EPCImp	2604	R	2	UInt32	kWh	Phase C Positive active energy
EPImp	2606	R	2	UInt32	kWh	Total Positive active energy
EPAExp	2608	R	2	UInt32	kWh	Phase A Reverse active energy
EPBExp	2610	R	2	UInt32	kWh	Phase B Reverse active energy
EPCExp	2612	R	2	UInt32	kWh	Phase C Reverse active energy
EPExp	2614	R	2	UInt32	kWh	Total Reverse active energy
Reactive energy- UInt32						
EQAImp	2616	R	2	UInt32	kVARh	Phase A Positive reactive

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
						energy
EQBImp	2618	R	2	UInt32	kVARh	Phase B Positive reactive energy
EQCImp	2620	R	2	UInt32	kVARh	Phase C Positive reactive energy
EQImp	2622	R	2	UInt32	kVARh	Total Positive reactive energy
EQAExp	2624	R	2	UInt32	kVARh	Phase A Reverse reactive energy
EQBExp	2626	R	2	UInt32	kVARh	Phase B Reverse reactive energy
EQCExp	2628	R	2	UInt32	kVARh	Phase C Reverse reactive energy
EQExp	2630	R	2	UInt32	kVARh	Total Reverse reactive energy
Apparent Energy-UInt32						
ESA	2632	R	2	UInt32	kVAh	Phase A Apparent Energy
ESB	2634	R	2	UInt32	kVAh	Phase B Apparent Energy
ESC	2636	R	2	UInt32	kVAh	Phase C Apparent Energy
ES	2638	R	2	UInt32	kVAh	Total Apparent Energy

1.9.12 Tariff Energy

Tariff energy types are Int64 and UInt32, whose unit size is different.

When the tariff of electricity energy reaches 1.0 x kWh, each tariff energy will be automatically reset to zero.

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Tariff Energy-Int64						
ET1	2700	R	4	Int64	Wh	Tariff 1 Active Energy
ET2	2704	R	4	Int64	Wh	Tariff 2 Active Energy
ET3	2708	R	4	Int64	Wh	Tariff 3 Active Energy

ET4	2712	R	4	Int64	Wh	Tariff 4 Active Energy
ET5	2716	R	4	Int64	Wh	Tariff 5 Active Energy
ET6	2720	R	4	Int64	Wh	Tariff 6 Active Energy
Tariff Energy-UInt32						
ET1	2750	R	2	UInt32	kWh	Tariff 1 Active Energy
ET2	2752	R	2	UInt32	kWh	Tariff 2 Active Energy
ET3	2754	R	2	UInt32	kWh	Tariff 3 Active Energy
ET4	2756	R	2	UInt32	kWh	Tariff 4 Active Energy
ET5	2758	R	2	UInt32	kWh	Tariff 5 Active Energy
ET6	2760	R	2	UInt32	kWh	Tariff 6 Active Energy

1.9.13 Demand parameters

Register alias	Register address (decimal)	R/W	Size	Type	Unit	Description
Basic parameters of demand						
DMDMethod	3000	R/WC	1	UInt16	-	Demand calculation method: 0= sliding type 1= fixed
DMD block	3001	R/RC	1	UInt16	Minute	Demand interval
PDMD Reset Time	3002	R	4	Date time	-	Peak demand reset date and time
Power demand						
PADemand	3020	R	2	Float32	kW	Current active power demand of phase A
PAPeak Demand	3022	R	2	Float32	kW	Peak demand of phase A active power
PAPeak Demand Date	3024	R	4	Date time	-	Occurrence time of peak demand of phase A active power
PBDemand	3028	R	2	Float32	kW	Current active power demand of phase B
PBPeak	3030	R	2	Float32	kW	Peak demand of phase B

Demand						active power
PBPeak Demand Date	3032	R	4	Date time	-	Occurrence time of peak demand of phase B active power
PCDemand	3036	R	2	Float32	kW	Current active power demand of phase C
PCPeak Demand	3038	R	2	Float32	kW	Peak demand of phase 3 active power
PCPeak Demand Date	3040	R	4	Date time	-	Occurrence time of peak demand of phase C active power
PSUMDemand	3044	R	2	Float32	kW	Current total active power demand
PSUMPeakDemand	3046	R	2	Float32	kW	Peak demand of total active power
PSUMPeakDemandDate	3048	R	4	Date time	-	Occurrence time of peak demand of total active power
QADemand	3052	R	2	Float32	kVar	Current reactive power demand of phase A
QAPeak Demand	3054	R	2	Float32	kVar	Peak demand of phase A reactive power
QAPeak Demand Date	3056	R	4	Date time	-	Occurrence time of peak demand of phase A reactive power
QBDemand	3060	R	2	Float32	kVar	Current reactive power demand of phase B
QBPeak Demand	3062	R	2	Float32	kVar	Peak demand of phase B reactive power
QBPeak Demand Date	3064	R	4	Date time	-	Occurrence time of peak demand of phase B reactive power
QCDemand	3068	R	2	Float32	kVar	Current reactive power demand of phase C
QCPeak Demand	3070	R	2	Float32	kVar	Peak demand of phase C reactive power
QCPeak Demand Date	3072	R	4	Date time	-	Occurrence time of peak demand of phase L3 reactive power

QSUMDemand	3076	R	2	Float32	kVar	Current total reactive power demand
QSUMPeakDemand	3078	R	2	Float32	kVar	Peak demand of total reactive power
QSUMPeakDemandDate	3080	R	4	Date time	-	Occurrence time of peak demand of total reactive power
SADemand	3084	R	2	Float32	kVa	Current apparent power demand of phase A
SAPeak Demand	3086	R	2	Float32	kVa	Peak demand of phase A apparent power
SAPeak Demand Date	3088	R	4	Date time	-	Occurrence time of peak demand of phase A apparent power
SBDemand	3092	R	2	Float32	kVa	Current apparent power demand of phase B
SBPeak Demand	3094	R	2	Float32	kVa	Peak demand of phase L2 apparent power
SBPeak Demand Date	3096	R	4	Date time	-	Occurrence time of peak demand of phase B apparent power
SCDemand	3100	R	2	Float32	kVa	Current apparent power demand of phase C
SCPeak Demand	3102	R	2	Float32	kVa	Peak demand of phase C apparent power
SCPeak Demand Date	3104	R	4	Date time	-	Occurrence time of peak demand of phase C apparent power
SSUMDemand	3108	R	2	Float32	kVa	Current total apparent power demand
SSUMPeakDemand	3110	R	2	Float32	kVa	Peak demand of total apparent power
SSUMPeakDemandDate	3112	R	4	Date time	-	Occurrence time of peak demand of total apparent power

1.9.14 Voltage and current harmonic register

Register name	Register address (decimal)	R/W	Size	Type	Unit	Description
Current harmonic percentage						
IATHD	4000	R	2	Float32	%	Phase A current total harmonic percentage
IBTHD	4002	R	2	Float32	%	Phase B current total harmonic percentage
ICTHD	4004	R	2	Float32	%	Phase C current total harmonic percentage
IATOHD	4006	R	2	Float32	%	Phase A current odd total harmonic percentage
IBTOHD	4008	R	2	Float32	%	Phaese B current odd total harmonic percentage
ICTOHD	4010	R	2	Float32	%	Phase C current odd total harmonic percentage
IATEHD	4012	R	2	Float32	%	Phase A current even total harmonic percentage
IBTEHD	4014	R	2	Float32	%	Phase B current even total harmonic percentage
ICTEHD	4016	R	2	Float32	%	Phsee C curremt even total harmonic percentage
IAHD1	4018	R	2	Float32	%	1st harmonic percentage of phase A current
IBHD1	4020	R	2	Float32	%	1st harmonic percentage of phase B current
ICHD1	4022	R	2	Float32	%	1st harmonic percentage of phase C current
…	4024-4311	…	…	…	…	The 2nd-49th harmonic percentage of ABC phase current
IAHD50	4312	R	2	Float32	%	The 50th harmonic percentage of phase A current
IBHD50	4314	R	2	Float32	%	The 50th harmonic percentage of phase B current
ICHD50	4316	R	2	Float32	%	The 50th harmonic percentage

						of phase C current
Current harmonic value						
IAHDV1	4400	R	2	Float32	A	Fundamental current value of phase A current
IBHDV1	4402	R	2	Float32	A	Fundamental current value of phase B current
ICHDV1	4404	R	2	Float32	A	Fundamental current value of phase C current
…	4406-4693	…	…	…	…	The 2nd - 49th harmonic current value of ABC phase current
IAHDV50	4694	R	2	Float32	A	The 50th harmonic current value of phase A current
IBHDV50	4696	R	2	Float32	A	The 50th harmonic current value of phase B current
ICHDV50	4698	R	2	Float32	A	The 50th harmonic current value of phase C current
Voltage harmonic percentage						
UATHD	5000	R	2	Float32	%	Phase A volage total harmonic percentage
UBTHD	5002	R	2	Float32	%	Phase B voltage total harmonic percentage
UCTHD	5004	R	2	Float32	%	Phase C voltage total harmonic percentage
UATOHD	5006	R	2	Float32	%	Phase A voltage odd total harmonic percentage
UBTOHD	5008	R	2	Float32	%	Phase B voltage odd total harmonic percentage
UCTOHD	5010	R	2	Float32	%	Phase C voltage odd total harmonic percentage
UATEHD	5012	R	2	Float32	%	Phase A voltage even total harmonic percentage
UBTEHD	5014	R	2	Float32	%	Phase B voltage even total harmonic percentage
UCTEHD	5016	R	2	Float32	%	Phase C voltage even total harmonic percentage
UAHDV1	5018	R	2	Float32	%	The 1st harmonic percentage of phase A voltage

UBHDV1	5020	R	2	Float32	%	The 1st harmonic percentage of phase B voltage
UCHDV1	5022	R	2	Float32	%	The 1st harmonic percentage of phase C voltage
…	5024-5311	…	…	…	…	The 2nd - 49th harmonic percentage of ABC phase voltage
UAHD50	5312	R	2	Float32	%	The 50th harmonic percentage of phase A voltage
UBHD50	5314	R	2	Float32	%	The 50th harmonic percentage of phase B voltage
UCHD50	5316	R	2	Float32	%	The 50th harmonic percentage of phase C voltage
Voltage harmonic value						
UAHDV1	5400	R	2	Float32	V	The 1st harmonic voltage value of phase A voltage
UBHDV1	5402	R	2	Float32	V	The 1st harmonic voltage value of phase B voltage
UCHDV1	5404	R	2	Float32	V	The 1st harmonic voltage value of phase C voltage
…	5406-5693	…	…	…	…	The 2nd - 49th harmonic voltage value of ABC phase voltage
UAHDV50	5694	R	2	Float32	V	The 50th harmonic voltage value of phase A voltage
UBHDV50	5696	R	2	Float32	V	The 50th harmonic voltage value of phase B voltage
UCHDV50	5698	R	2	Float32	V	The 50th harmonic voltage value of phase C voltage

1.9.15 Max. value and Min. value

Register name	Register address (decimal)	R/W	Size	Type	Unit	Description
Current max / min						
IA Max	6000	R	2	Float32	A	Phase A Maximum current
IB Max	6002	R	2	Float32	A	Phase B Maximum current
IC Max	6004	R	2	Float32	A	Phase C Maximum current

IAVG Max	6006	R	2	Float32	A	Maximum three-phase average current
IN Max	6008	R	2	Float32	A	Phase N Maximum current
IA Min	6010	R	2	Float32	A	Phase A Minimum current
IB Min	6012	R	2	Float32	A	Phase B Minimum current
IC Min	6014	R	2	Float32	A	Phase C Minimum current
IAVGMin	6016	R	2	Float32	A	Minimum three-phase average current
IN Min	6018	R	2	Float32	A	Phase N Minimum current
Voltage max / min						
UA Max	6020	R	2	Float32	V	UA-UN Maximum phase voltage
UB Max	6022	R	2	Float32	V	UB-UN Maximum phase voltage
UC Max	6024	R	2	Float32	V	UC-UN Maximum phase voltage
Phase UAVGMax	6026	R	2	Float32	V	Maximum value of average value of three-phase phase voltage
UA Min	6030	R	2	Float32	V	UA-UN Minimum phase voltage
UB Min	6032	R	2	Float32	V	UB-UN Minimum phase voltage
UC Min	6034	R	2	Float32	V	UC-UN Minimum phase voltage
UAVGMin	6036	R	2	Float32	V	Minimum of average value of three-phase phase voltage
UAB Max	6040	R	2	Float32	V	UA-UB Maximum wire voltage
UBC Max	6042	R	2	Float32	V	UB-UC Maximum wire voltage
UCA Max	6044	R	2	Float32	V	UC-UA Maximum wire voltage
LineUAVG Max	6046	R	2	Float32	V	Maximum value of average value of three-phase phase voltage
UAB Min	6050	R	2	Float32	V	UA-UB Minimum phase voltage
UBC Min	6052	R	2	Float32	V	UB-UC Minimum phase voltage
UCA Min	6054	R	2	Float32	V	UC-UA Minimum phase voltage
LineUAVG Min	6056	R	2	Float32	V	Minimum of average value of three-phase phase voltage
Maximum / minimum power						
PA Max	6060	R	2	Float32	kW	Maximum active power of phase A
PB Max	6062	R	2	Float32	kW	Maximum active power of phase B
PC Max	6064	R	2	Float32	kW	Maximum active power of phase C

PSUMMax	6066	R	2	Float32	kW	Maximum value of three-phase total active power
PA Min	6070	R	2	Float32	kW	Minimum active power of phase A
PB Min	6072	R	2	Float32	kW	Minimum active power of phase B
PC Min	6074	R	2	Float32	kW	Minimum active power of phase C
PSUMMin	6076	R	2	Float32	kW	Minimum value of three-phase total active power
Reactive Power Max / min						
QA Max	6080	R	2	Float32	kVar	Maximum value of phase A reactive power
QB Max	6082	R	2	Float32	kVar	Maximum value of phase B reactive powe
QC Max	6084	R	2	Float32	kVar	Maximum value of phase C reactive powe
QSUMMax	6086	R	2	Float32	kVar	Maximum value of three-phase total reactive power
QA Min	6090	R	2	Float32	kVar	Minimum value of phase A reactive power
QB Min	6092	R	2	Float32	kVar	Minimum value of phase B reactive power
QC Min	6094	R	2	Float32	kVar	Minimum value of phase C reactive power
QSUMMin	6096	R	2	Float32	kVar	Minimum value of three-phase total reactive power
Apparent power max / min						
SA Max	6100	R	2	Float32	kVa	Maximum apparent power of phase A
SB Max	6102	R	2	Float32	kVa	Maximum apparent power of phase B
SC Max	6104	R	2	Float32	kVa	Maximum apparent power of phase C
SSUMMax	6106	R	2	Float32	kVa	Maximum three-phase totalapparent power
SA Min	6110	R	2	Float32	kVa	Minimum apparent power of phase A
SB Min	6112	R	2	Float32	kVa	Minimum apparent power of phase B

SC Min	6114	R	2	Float32	kVa	Minimum apparent power of phase C
SSUMMin	6116	R	2	Float32	kVa	Minimum three-phase total apparent power

1.9.16 Unbalancedness

Register name	Register address (decimal)	R/W	Size	Type	Unit	Description
Current negative sequence unbalancedness	7000	R	2	Float32	%	Current negative sequence unbalancedness
Current zero sequence unbalancedness	7002	R	2	Float32	%	Current zero sequence unbalancedness
Voltage negative sequence unbalancedness	7004	R	2	Float32	%	Voltage negative sequence unbalancedness
Voltage zero sequence unbalancedness	7006	R	2	Float32	%	Voltage zero sequence unbalancedness

1.9.17 Current K-factor and crest factor register

Register name	Register address (decimal)	R/W	Size	Type	Unit	Description
Current K factor						
KFIA	8000	R	2	Float32	-	Current K factor of phase A
KFIB	8002	R	2	Float32	-	Current K factor of phase B
KFIC	8004	R	2	Float32	-	Current K factor of phase C

1.9.18 Voltage and current angle register

Register name	Register address (decimal)	R/W	Size	Type	Unit	Description
Angle of voltages						
UA	8100	R	2	Float32		Angle of phase A voltage

UB	8102	R	2	Float32		Angle of phase B voltage
UC	8104	R	2	Float32		Angle of phase C voltage
Angle of currents						
IA	8106	R	2	Float32		Angle of phase A current
IB	8108	R	2	Float32		Angle of phase B current
IC	8110	R	2	Float32		Angle of phase C current
Angle between voltage and current						
UIA	8112	R	2	Float32		Angle between voltage and current of phase A
UIB	8114	R	2	Float32		Angle between voltage and current of phase B
UIC	8116	R	2	Float32		Angle between voltage and current of phase C

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